

← Back

Range Trees

Practice Quiz • 20 min • 5 total points

≡ Hide menu

Introduction to Range Searching

1D Range Searching

KD Trees

Queries in KD-Trees

Range Trees

Range Trees: Extensions

Video: Range Trees: Extensions

Practice Quiz: Range Trees

Programming: Orthogonal Range Searching

Wrap-Up Quiz

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Range Trees

Go to next item

1. In some applications one is interested only in the number of points that lie in range rather than in reporting all of them. Such queries are often referred to as **range counting queries**. In this case one would like to avoid having an additive term of $O(k)$ in the query time. Consider the 1-dimensional range tree in the image below, we can perform a range counting query in $O(\log n)$ time by storing a **leaf count** number in all of the internal nodes, which sequence ordered by the node labels corresponds to the numbers we should store in the internal nodes?

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Your grade 100%

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🔴 9 5 4 3 2 3 2 2

🔴 3 7 9 7 5 3 5 3 3

🔴 1 0 0 1 0 1 1 0

🔴 Correct

Correct, if for each internal node we store the number of leafs in that subtree we can do range counting queries in $O(\log n)$ time.

2. What happens to the query time, construction time and storage requirements of range trees when we apply fractional cascading?

☐ We need more storage.

☒ The storage requirement stays the same.

☐ Correct

Correct.

☐ We need less storage.

☐ The construction time is increased.

☒ The construction time stays the same.

☐ Correct

Correct.

☐ The construction time is decreased.

☐ The query time is increased.

☐ The query time stays the same.

☒ The query time is decreased.

☐ Correct

Correct.

3. Let S_1 be a set of n disjoint horizontal line segments and let S_2 be a set of m disjoint vertical line segments. A plane-sweep algorithm that outputs the number of intersections in $S_1 \cup S_2$ takes $O((n + m) \log(n + m))$ time. How is this running time achieved?

☐ We build a 4-dimensional range tree on the coordinates of the horizontal line segments, and then perform range-counting queries corresponding to the vertical line segments.

☐ We run the line-segment intersection algorithm from the first module, but sweep diagonally to avoid degenerate cases. Everytime we encounter an intersection, we increase a counter, which we then return at the end.

☒ As a status we have the currently intersected vertical line segments in a binary search tree. For every horizontal line segment we have an event at which we perform a 1-dimensional range counting query on the binary search tree.

☐ Correct

Correct.

4. Consider the point set and range tree shown above. Every point p_i has x -coordinate x_i and y -coordinate y_i . The range tree on the right first splits based on x -coordinate. Which elements are in the associated data structure (indicated by the triangle) of the blue node?

☒ $p_9, p_{10}, p_{11}, p_{12}$

☐ p_1, p_2, p_3, p_4

☐ $y_9, y_{10}, y_{11}, y_{12}$

☐ y_1, y_2, y_3, y_4

☐ Correct

Incorrect, the associated data structure stores y -coordinates.

5. Consider the range tree and point set in the image above. Suppose we want to perform the query indicated by the blue square. In which internal nodes do we perform a search in the associated data structure (indicated by triangles)?

☐ 1

☐ 2,3

☒ 6,11,14

☐ 6, 11, 12, 13, 14

☐ Correct

Correct, the x -range of the query rectangle fully contains regions 6, 11 and 14.

https://www.coursera.org/learn/geometric-algorithms/quiz/5JfG/range-trees/attempt/redirectToCourse=true

1/1